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RESEARCH ARTICLE

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A palaeomagnetic study of the Middle Permian and Middle Triassic limestones from Shan State, Myanmar: Implications for collision of the Sibumasu Terrane and Indochina Terrane

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We report herein a new palaeomagnetic study on the Middle Permian and Middle Triassic limestones from Shan State of eastern Myanmar, aiming to resolve the spatial location of the Sibumasu Terrane during the collision of tectonic blocks in Southeast Asia and to constrain the timing of convergence between the Sibumasu Terrane and Indochina Terrane. Based on the rock magnetic and palaeomagnetic experiments, three stable characteristic remnant magnetizations are recognized from limestone specimens of Middle Permian and Middle Triassic age. The characteristic Component C of the Middle Permian limestone passed the fold test at 95% confidence level, indicative of a pre-folding origin. The mean direction of this component is D = 10.7°, I = 17.9° (κ = 130.3, α_{95} = 10.8°) before and D = 13.7°, I = −25.3° (κ = 1894.9, α_{95} = 2.8°) after tilt correction, corresponding to a palaeomagnetic pole at 254.1°E, 53.2°N (A_{95} = 1.7°). This suggests that the Sibumasu Terrane was placed at a palaeolatitude of ~13.0°S during the Middle Permian. The Component C separated from the Middle Triassic samples also passed the fold test with a mean direction of D = 207.3°, I = −8.3° (κ = 12.9, α_{95} = 6.2°) before tilt correction and D = 205.6°, I = −33.5° (κ = 27.1, α_{95} = 4.2°) after tilt correction, corresponding to a palaeomagnetic pole at 187.3°E, 65.8°N (dp/dm = 2.7°/4.8°). This implies that the Sibumasu Terrane was located at a palaeolatitude of ~18.3°N during the Middle Triassic. Based upon the detailed palaeomagnetic analysis, this study supports the view that the collision of the Sibumasu Terrane and Indochina Terrance occurred during or before the Middle–Late Triassic times.

KEYWORDS

eastern Palaeotethys, Middle Permian, Middle Triassic, palaeomagnetism, Shan State, Sibumasu Terrane, Southeast Asia

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1 | INTRODUCTION

Southeast Asia is made up of a giant jigsaw puzzle of continental terranes of Gondwanan origin (e.g., Indochina, Sibumasu, and west Burma) that amalgamated during the Mesozoic (Metcalfe, 1996; Şengör, 1984). The tectonic evolution of the eastern Palaeotethys that separates the Sibumasu Terrane and Indochina Terrane is essential for understanding the geological history of Southeast Asia. Despite significant research efforts in sedimentology (Liu, Feng, Fang, Jia, & He, 1993), palaeontology (Shi & Archbold, 1998; Ueno, 2003), and geochronology (Dong et al., 2013; Jian et al., 2009; Y. Wang et al., 2010) over the past decades, the closure timing and location of eastern Palaeotethys are still matters of debate largely due to lack of quantitative geographic constraints, such as high-resolution palaeomagnetic results for the terranes and blocks of Southeast Asia (Collins, 2003; Metcalfe, 2013; Scotese, 2004; Stampfli, Hochard, Vérard, Wilhem, & vonRaumer, 2013). Some authors suggested that the eastern Palaeotethys closed in the Early–Middle Triassic (Carter, Roques, Bristow, & Kinny, 2001; Y. Wang et al., 2016) or in the Middle to Late Triassic (Sone & Metcalfe, 2008), while others argued that the closure occurred during the Early to Middle Jurassic (Collins, 2003; Metcalfe, 2017; Scotese, 2004).

Due to intensive deformation and significant magmatism caused by the India–Asia collision (Meng et al., 2017, 2018; Otofuji et al., 2012; Tapponnier, Peltzer, Dain, Armijo, & Cobbold, 1982), this area has experienced pervasive remagnetization with very few available Late Palaeozoic to Early Mesozoic palaeomagnetic data (Ali, Cheung, Aitchison, & Sun, 2013; Chan, Wang, & Wu, 1984; Fang, Van der Voo, & Liang, 1989; K. Huang & Opdyke, 1991) that meets the palaeomagnetic reliability criteria (e.g., Van der Voo, 1990). Moreover, the available palaeomagnetic data are not well distributed across the region with most data being from the Baoshan Block and the Simao Block in the northern extension of the Sibumasu Terrane and Indochina Terrane (e.g., Huang & Opdyke, 2016; Otofuji et al., 2012; Zhao, Huang, Yan, & Zhang, 2015). This largely prevents accurate constraints on the palaeoposition of the stripped blocks.

To resolve this issue, we conducted a detailed palaeomagnetic study on the Middle Permian and Middle Triassic limestones from the Shan State of eastern Myanmar in the central Sibumasu Terrane, aiming to provide accurate palaeomagnetic constraints on the palaeogeography of the Sibumasu Terrane during the Middle Permian and Middle Triassic and to resolve the timing and location of the Sibumasu–Indochina collision.

2 | GEOLOGICAL SETTING AND SAMPLING

The Sibumasu Terrane is bounded by the N–S-striking Sagaing Fault and Mogok metamorphic belt to the west (Figure 1; Aung, 2009; Barber, Ridd, & Crow, 2011; Burrett et al., 2014; Metcalfe, 2011; Gardiner, Searle, Robb, & Morley, 2015; Morley, 2012; Sone & Metcalfe, 2008) and by the Chiang Mai–Chiang Rai Suture to the east (A. H. G. Mitchell, 1977; Metcalfe, 2017; Ridd, 2016). It is thought to have been part of the Gondwana supercontinent before the middle Early Permian (Ali et al., 2013; Burrett et al., 2014; Burrett, Long, & Stait, 1990; H. Huang, Jin, Shi, & Yang, 2009; Jin, 2002; Metcalfe, 1996; Y. Wang, Xing, et al., 2013; X. D. Wang & Sugiyama, 2002) and have accreted into the Indochina Terrane during the early Mesozoic (Metcalfe, 2013; Y. Wang et al., 2010).

The Middle Permian to Middle or Upper Triassic carbonates were broadly deposited in the Shan State (Garson, Amos, & Mitchell, 1976; A. H. G. Mitchell, 1977; Oo, Hlaing, & Htay, 2002). They are unconformably underlain by the Lower Palaeozoic to Devonian stratigraphic sequences (Amos, 1975) and overlain by the Late Triassic–Early Jurassic siliciclastic units (A. H. G. Mitchell et al., 2004, 2007; Mitchell, Chung, Oo, Lin, & Hung, 2012; Ridd & Watkinson, 2013). These carbonates are classified as the Plateau Limestone Group that comprises the Permian Thitsipin Limestone Formation, the Late Permian–Early Triassic Nwabangyi Dolomite Formation, and the Middle Triassic Natteik Limestone Formation (Amos, 1975; Garson et al., 1976; A. H. G. Mitchell, 1977). In the southern Shan State, Carboniferous strata overlain by the Permian and Triassic carbonates are also observed (Metcalfe & Aung, 2014 and references therein).

Palaeomagnetic samples were systematically collected from the Middle Permian and Middle Triassic carbonates in the Taunggyi, Aungban, Pindya, Ywangan, Yakainggyi, and Kyause areas of the west Shan Plateau (Figures 1 and 2). During the field sampling, nine to 13 individual cores were drilled at each site. In total, 415 samples from 32 sites of the Middle Permian carbonates and 152 samples from 15 sites of the Middle Triassic carbonates were collected using a portable gasoline-powered drill and a sun compass. The sampling sites are marked in the geological map of the western Shan State (Figure 2).

3 | LABORATORY METHODS

3.1 | Rock magnetism

To identify the magnetic minerals within the samples, rock magnetic experiments, including acquisition of isothermal magnetization (IRM), back-field demagnetization of saturation IRM (SIRM), thermal demagnetization of composition IRMs (Lowrie, 1990), and magnetic susceptibility versus temperature (κ-T), were conducted for 12 pilot samples. The acquisition of isothermal remanent magnetization (IRM) and back-field demagnetization of saturation isothermal remanent magnetization (SIRM) were carried out using a Micromag 3900 alternating gradient magnetometer. The κ-T experiments were performed on a KLY-3S instrument by heating up the sample from room temperature to 700°C and subsequently cooling back to room temperature under an Argon gas circulation circumstance. For the thermal demagnetization of the composition IRMs, the samples were magnetized using a pulse magnetizer (2G-660) with direct current (DC) magnetic field of 2.7, 0.4, and 0.1 T being applied to the Z-axis, Y-axis, and X-axis successively. Thermal demagnetization up to 680°C was performed with a TD-48 thermal demagnetizer, and remanent magnetizations were measured using a 2G-755 cryogenic magnetometer. All the

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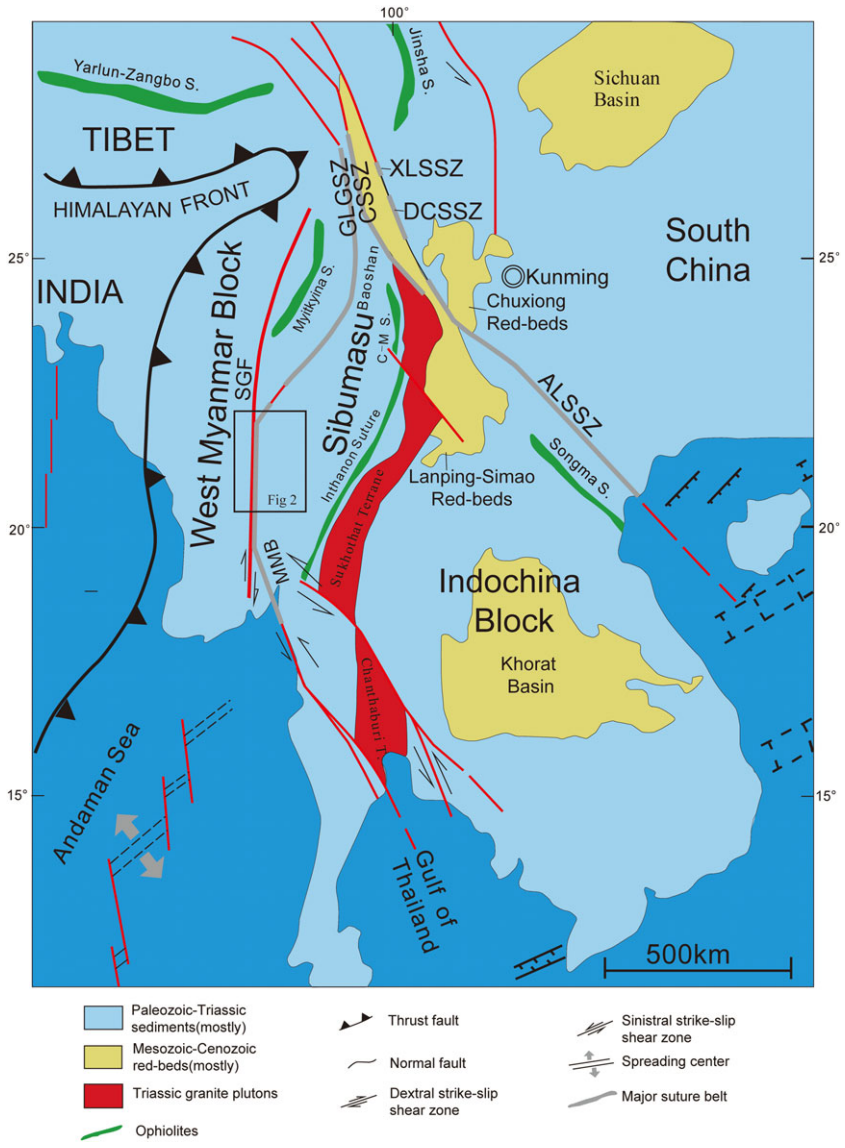


FIGURE 1 Schematic tectonic map of Southeast Asia (after Leloup et al., 1995; Metcalfe, 2013) showing the major terranes and sutures. C–M S: Changning–Menglian Suture. Xuelongshan Shear Zone, XLGSZ, DCSSZ (Diancangshan Shear Zone), CSSZ (Chongshan Shear Zone), SGF (Sagaing Fault), MMB (Mogok metamorphic zone) [Colour figure can be viewed at wileyonlinelibrary.com]

rock magnetism measurements and palaeomagnetic experiments were conducted at the Paleomagnetism and Geochronology Laboratory of Institute of Geology and Geophysics, Chinese Academy of Sciences.

3.2 | Palaeomagnetism

Cylindrical specimens with length of ~2.0–2.2 cm were cropped from field core samples. Following the evaluation of rock magnetic behaviours from pilot samples, all specimens were subjected to progressive thermal demagnetization or an integrated approach comprising thermal demagnetization to ~250°C and followed alternating field (AF) demagnetization up to 80 mT. The thermal demagnetization was carried out in the PGL-TD-100 thermal demagnetizer with temperature intervals of 20–100°C to 500°C and subsequent intervals of 10°C as the maximum unblocking temperatures of the remanence carriers were approached. Magnetizations were measured using a 2G-RAPID cryogenic magnetometer. The magnetometers and demagnetizers are installed in a magnetic-shielded room with the field being minimized to less than 300 nT. Demagnetization results were analysed using the stereographic projections and orthogonal diagrams (Zijderveld, 1967) with the latter being used to resolve the component structures. The equivalent component directions were determined using the principal component analysis (PCA) method (Kirschvink, 1980) or great circle fitting (GC; P. L. McFadden & McElhinny, 1988), while the interval–mean were estimated using the standard Fisherian method (Fisher, 1953) or a combination of remagnetization circles and direction observations (P. L. McFadden & McElhinny, 1988).

4 | RESULTS

4.1 | Rock magnetic results

According to the experimental results of rock magnetism, the samples are characterized by three categories. Most samples show a complete saturation at ~0.3 T with remanence coercivity (Hcr) less than 0.1 T (Figure 3b), while the IRM component analyses only identify one soft

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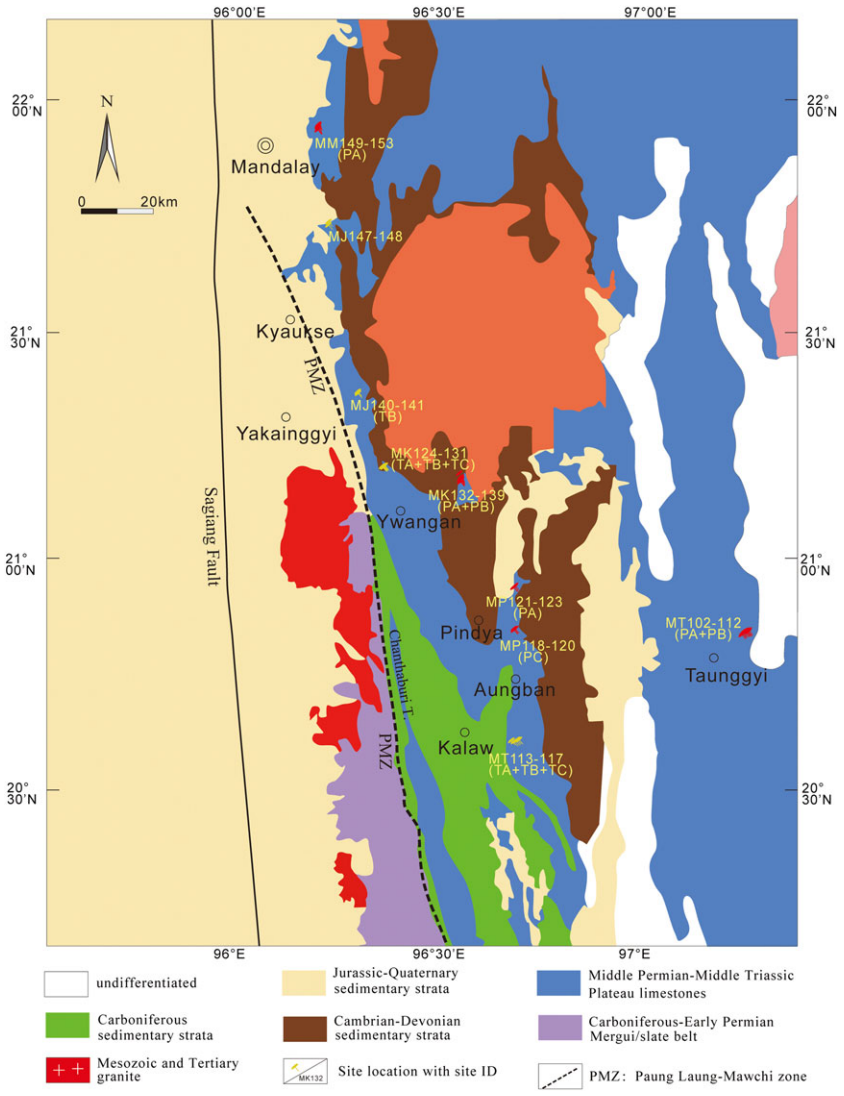


FIGURE 2 Simplified geologic map of the west part of Shan State in the northeast of Myanmar (after Myanmar Geosciences Society, 2011; A. Mitchell et al., 2012). The red and yellow nail symbol represent the sites location of Middle Permian and Middle Triassic [Colour figure can be viewed at wileyonlinelibrary.com]

component (Figure 3c; Kruiver, Dekkers, & Heslop, 2001). Similarly, demagnetization of composite IRMs indicates that the soft and medium components unblock completely below 450°C (Figure 3a). Furthermore, the κ-T curve shows that the magnetic susceptibility drops markedly to zero at approximately 580°C (Figure 3d), which is indicative of the presence of magnetite.

Several Middle Permian limestone samples and a number of Middle Triassic limestone samples are characterized by a rapid increase in IRM intensity before 0.2 T, and an additional increase up to 1.5 T, the remanence coercivity generally lies at approximately 0.2 T (Figure 3f). Further IRM component analysis exhibits high and low coercivity components (Figure 3g), suggesting the presence of both low and high coercivity magnetic minerals. Demagnetization of the composite IRMs reveals that the soft and hard components experience a rapid decrease of remanence at around 100°C and completely unblock at 620–660°C, indicative of the presence of goethite and haematite (Figure 3e).

Additionally, some Middle Triassic limestones from the section at the south of Aungban town are characterized by more complicated rock magnetic behaviours. The dominative soft component unblocking at 500–525°C suggests the presence of coarse grain-size (PSD) magnetite or Ti-poor titanomagnetite and goethite, the soft and hard components show a significant decrease at ~100°C (Figure 3i) indicating the existence of goethite. Note more precise mineral composition requires further experimental works, but beyond the scope of this study.

Collectively, the main magnetic carriers for the Middle Permian and Middle Triassic limestones are dominated by magnetite with slight outliers of a combination of magnetite, haematite, and goethite.

4.2 | Palaeomagnetic results

4.2.1 | Middle Permian limestones

In general, a soft viscous component or a low-temperature component is isolated from 168 out of 270 demagnetized specimens below ~15 mT or 150–200°C (Figure 4). This component has an in situ mean direction of D = 0.6°, I = 35.4° (κ = 19.8, α_{95} = 2.5°) with significant deterioration of data grouping after tilt correction (κ_s/κ_g = 0.34), suggesting a negative fold test (Figure 5a,b). This component is thought

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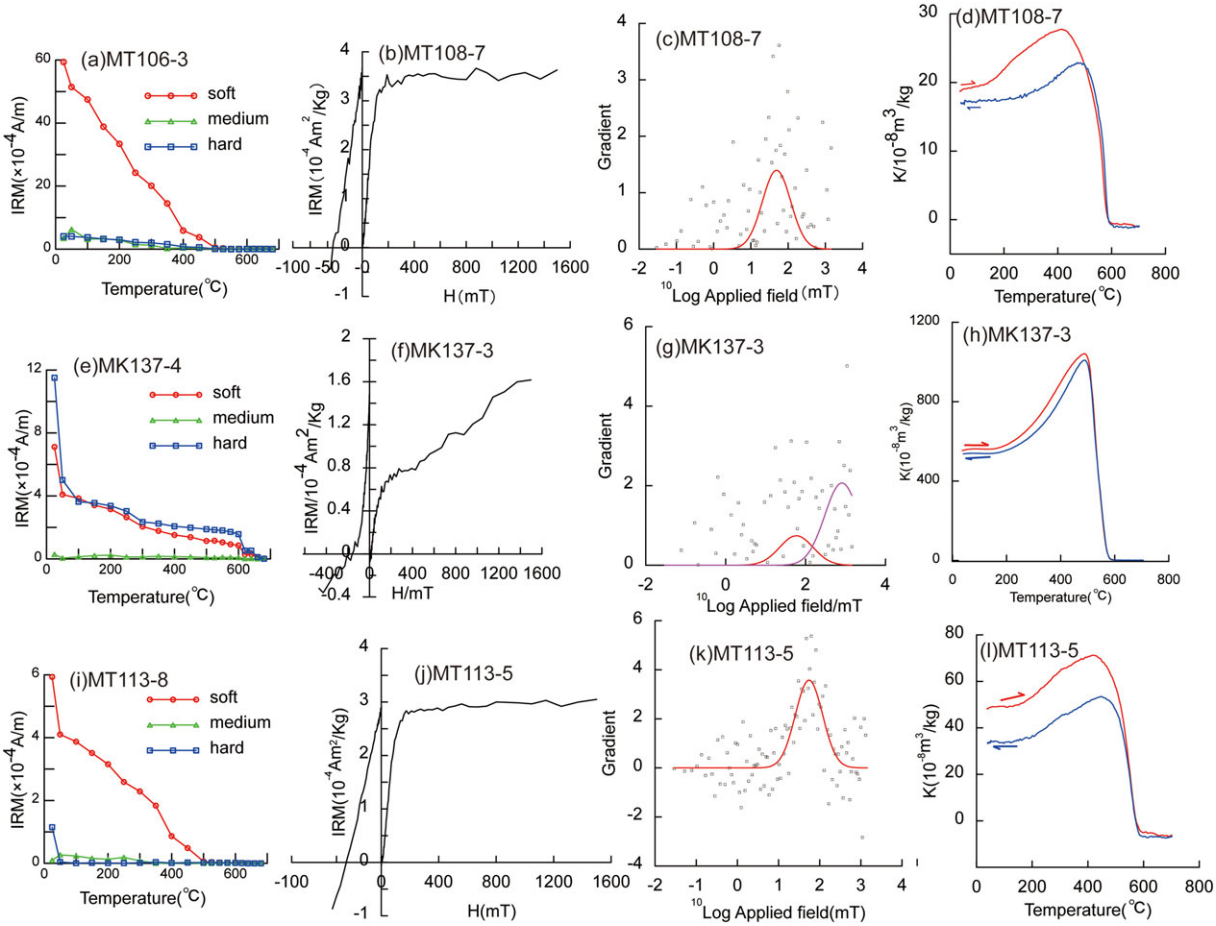


FIGURE 3 Rock magnetic results for representative samples of the Middle Permian and Middle Triassic rocks from Shan State: (a, e, i) composite IRMs thermal demagnetization curves; (b, f, j) acquisition curves of isothermal remanent magnetization (IRM) and back-field demagnetization curves of saturated IRM; (c, g, k) examples of IRM component analysis (open circles: raw data, red line: Comp.1, green line: Comp. 2); (d, k, l) susceptibility versus temperature curves (red and blue line indicate heating and cooling curves) [Colour figure can be viewed at wileyonlinelibrary.com]

to result from the recent overprinting in the light of conformity to the present geomagnetic field (D = −0.8°, I = 30.4°; Thébault et al., 2015).

Following the removal, 142 specimens yield a well-defined characteristic remanence between 5–35 to 20–80 mT and 50–400°C to 200–620°C (Figure 4a,c,d). Twenty-nine of the remaining specimens exhibit high-temperature or high-field demagnetization trajectory following a great circle (Figure 4b), allowing characteristic remanence (ChRM) to be isolated using the method of remagnetization great circle (P. L. McFadden & McElhinny, 1988).

The high-temperature/field ChRMs can be divided into three components based on their distributions (Figure 6). First, Component A (from nine sites) yields a mean direction of D = 0.2°, I = 37.8° (κ = 34.5, α_{95} = 8.9°) before tilt correction and D = 10.3°, I = 37.3° (κ = 11.5, α_{95} = 15.9°) after tilt correction (Table 1 and Figure 5a,b). Further analysis by the stepwise fold test of Watson and Enkin (1993) highlights an optimal concentration at −15% unfolding, indicating a post-folding origin for the ChRM (Figure 6c).

Second, Component B (from nine sites) is characterized by a mean direction of D = 5.9°, I = 31.3° (κ = 25.5, α_{95} = 10.4°) before tilt correction and D = 13.5°, I = 16.7° (κ = 67.6, α_{95} = 6.3°) after tilt correction (Table 1 and Figure 6d,e). Further analysis using the Watson and Enkin (1993) test exhibits an optimal concentration at 140% unfolding, indicative of a syn-folding origin for the ChRM (Figure 6f).

Third, Component C (from three sites) presents a mean direction of D = 10.7°, I = 17.9° (κ = 130.3, α_{95} = 10.8°) before tilt correction and D = 13.7°, I = −25.3° (κ = 1894.9, α_{95} = 2.8°) after tilt correction (Table 1 and Figure 6g,h). Further analysis using the Watson and Enkin (1993) test gives an optimal concentration at 108% unfolding, suggesting its pre-folding origin (Figure 6i).

4.2.2 | Middle Triassic limestones

A low-temperature component can be isolated from 89 out of 110 Triassic demagnetized specimens below 250–300°C. This component exhibits an in situ mean direction of D = 350.1°, I = 31.6° (κ = 46.8, α_{95} = 5.7°) with obvious decrease in data grouping precision after tilt correction (κ_s/κ_g = 0.44), as illustrated by a negative fold test (Figure 5c,d). We interpret this low-temperature component to be the result of a recent overprinting in the light of conformity to the PGF.

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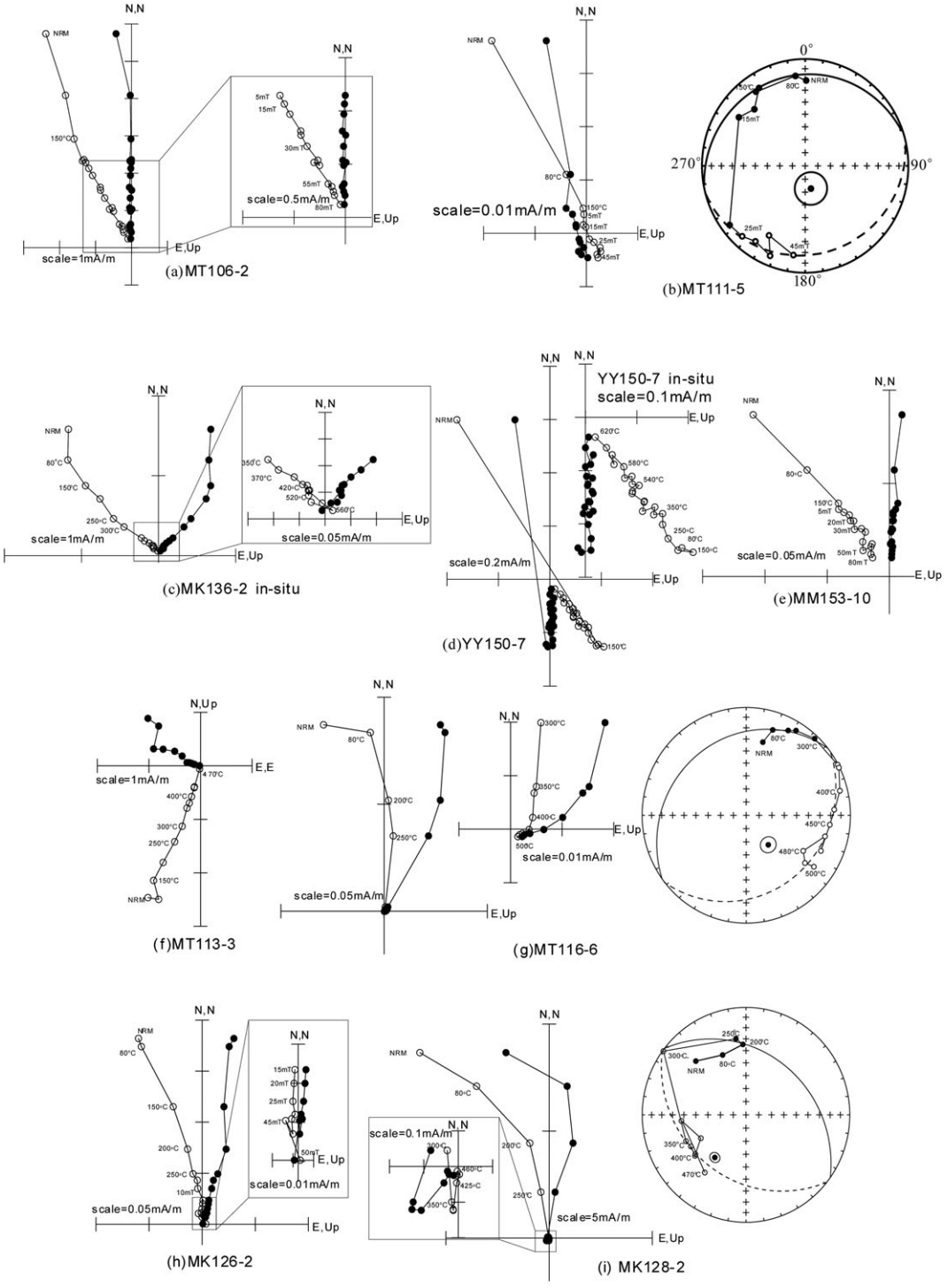


FIGURE 4 Representative orthogonal plots of thermal demagnetization for the samples from Middle Permian and Middle Triassic limestones from Shan State. Orthogonal directions (Zijderveld, 1967) are plotted in situ; while stereographic directions are plotted in stratigraphic coordinates. Solid/open symbols represent endpoints projected onto horizontal or vertical planes, respectively

The high-temperature or high-field ChRM can be isolated from 98 out of 110 specimens within the temperature interval of 300–500°C or field interval of 30–80 mT. Within this range, there are 51 specimens characterized by a distinct distribution of NRM endpoints along a great circle, indicating that ChRM is likely to have been obliterated (Figure 4g,i). According to the distribution of high-temperature or high-field ChRMs on the equal-area projections, the ChRM is categorized into three groups (Table S1) at specimen level. First, Component A is composed of ChRM isolated from 15 specimens with the mean direction of D = 356.4°, I = 13.5° (κ = 16.3, α_{95} = 9.8°) before tilt correction and D = 351.4°, I = 15.4° (κ = 14.5, α_{95} = 10.4°) after tilt correction (Figure 7a,b). Further analysis using the test of Watson

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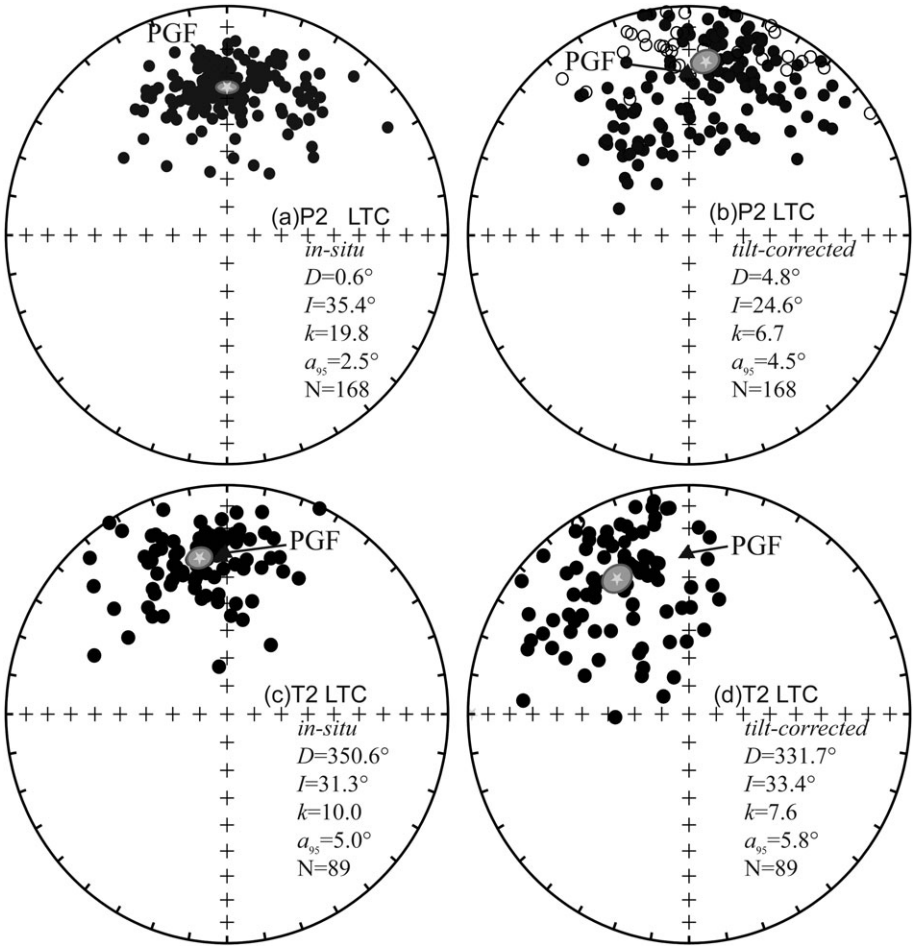


FIGURE 5 Equal-area projections of sample LTC directions of Middle Permian and Middle Triassic rocks from Shan State. Open (close) circles represent directions on the upper (lower) hemisphere. Grey pentagram and the ellipses denote the mean direction and its 95% confidence circle around it, black triangle denote the present geomagnetic field (PGF) direction at the sampling locality

and Enkin (1993) presents an optimal concentration at 13.7% unfolding, indicative of a post-folding origin for the ChRM (Figure 7c).

Second, Component B is made up of 24 specimens, its mean direction is D = 122.0°, I = 18.3° (κ = 15.5, α_{95} = 7.8°) before tilt correction and D = 134.1°, I = 42.1° (κ = 21.9, α_{95} = 6.1°) after tilt correction (Figure 7d,e). Further analysis using the test of Watson and Enkin (1993) highlights an optimal concentration at 71.5% unfolding, indicating a syn-folding origin (Figure 7f). The mean direction at 71.5% unfolding is D = 128.5°, I = 36.2° (κ = 23.99, α_{95} = 6.2°), which corresponds to a pole at 150.8°E, 25.2°S (dp/dm = 7.2°/4.2°).

Third, the mean direction of Component C, calculated based on ChRMs from 44 specimens, is D = 207.3°, I = −8.3° (κ = 12.9, α_{95} = 6.2°) before tilt correction and D = 205.6°, I = −33.5° (κ = 27.1, α_{95} = 4.2°) after tilt correction (Figure 7h,i), and the test of Watson and Enkin (1993) produces an optimal concentration at 102.0% unfolding, suggesting a pre-folding origin (Figure 7j). The precision parameter ratio of this component before and after tilt correction is 2.10 (McElhinny, 1964), suggesting a positive fold test result. The corresponding pole, located at 187.3°E, 65.8°N (dp/dm = 2.7°/4.8°), represents a palaeolatitude of ~18.3°N for the Shan State (reference site: 96.5°E, 20.5°N).

5 | DISCUSSION

5.1 | Timing of the remanence acquisition

5.1.1 | Middle Permian

Three stable components were isolated from the Middle Permian specimens, among which the Component C passed the fold test and indicates a pre-folding origin. As the regional deformation occurred during the Jurassic–Cretaceous and Cenozoic (K. Huang & Opdyke, 1991; Liao, Chen, Wei, & Li, 2003), we infer that this component was acquired before the Jurassic and probably during or shortly after the sedimentation. It indicates a palaeolatitude of ~13.0 ± 1.7°S for the Sibumasu Terrane (reference site: 96.5°E, 20.5°N).

As the Sibumusu Terrane generally moved northward from the Gondwana margin in the middle latitudes of the Southern Hemisphere to the low-middle latitudes of the Northern Hemisphere during the Early Permian to Late Triassic (Metcalfe, 2011), the characteristic Component B with low latitude of ~8.5°N (reference site: 96.5°E, 20.5°N) is further interpreted to be obtained later than that of the

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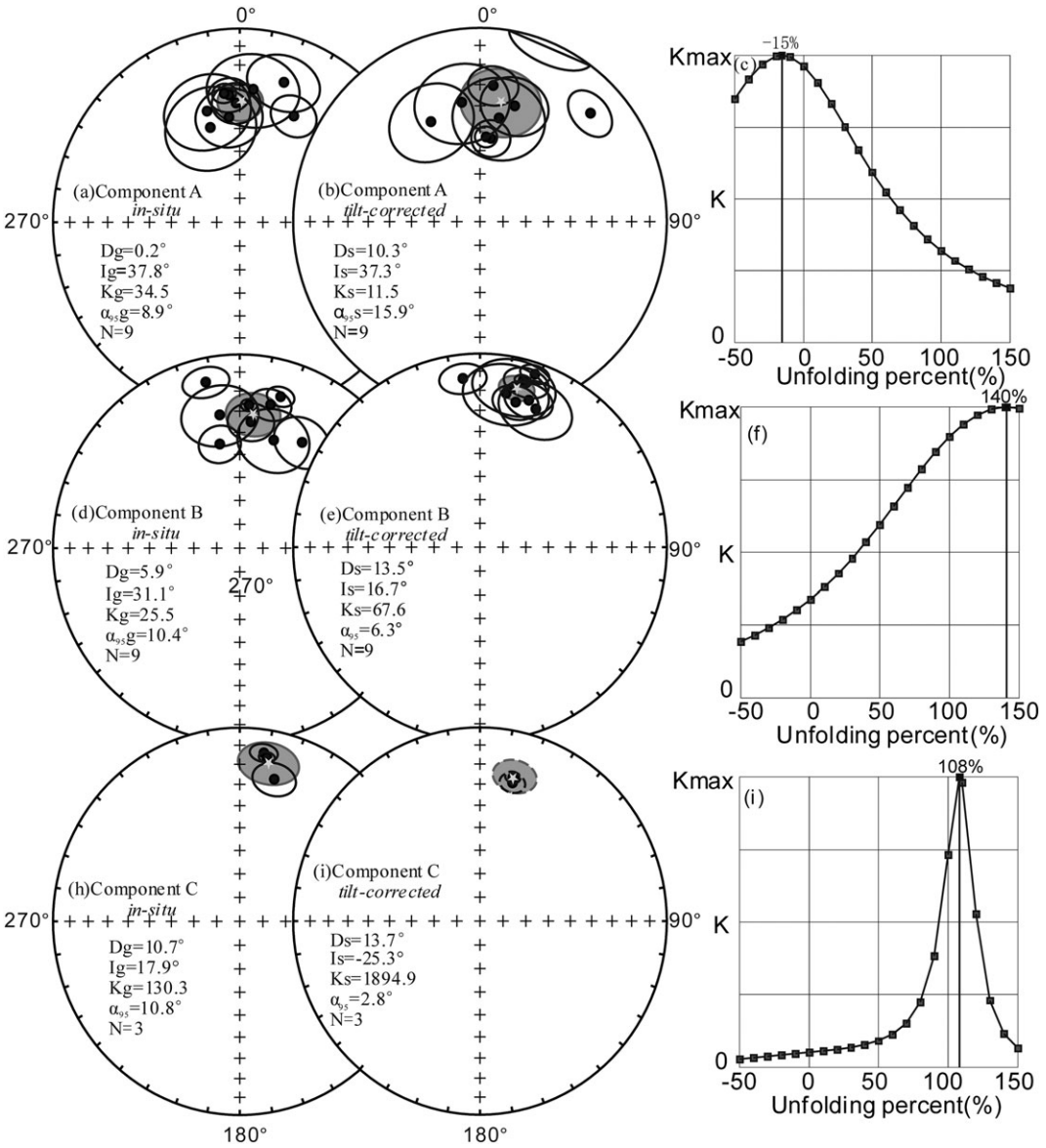


FIGURE 6 Equal-area projections of the site-mean observations of the Middle Permian limestones from Shan State (a, d, h) before and (b, e, i) after tilt correction, (c, f, j) incremental unfolding analysis using Watson and Enkin (1993) for tilt correction, for Components A, B, C, respectively

Component C. The accurate magnetization timing of the Component B requires additional investigation.

5.1.2 | Middle Triassic

Three stable components were isolated from the Middle Triassic specimens, in which Components A and B yield a negative fold test (Figure 7; Watson & Enkin, 1993) with optimal concentrations at ~6.1% and ~68.0% unfolding level, corresponding to post-folding and syn-folding origins, respectively. This indicates that the Component A is of post-folding origin. Additionally, the corresponding palaeomagnetic pole of the Component B appears to be very close to the palaeomagnetic pole determined from the mafic dykes (~50–30 Ma) of Tengchong (Kornfeld, Eckert, Appel, Ratschbacher, Sonntag, et al., 2014), implying that this component was obtained during the India–Asia collision.

An optimal concentration is acquired at ~102.0% unfolding for the Component C that passed the fold test of McElhinny (1964) at the 95% confidence level. This suggests that the Component C is attained before the regional folding during the Late Jurassic–Cretaceous and probably represents the primary magnetization phase during the rock formation. Given these facts, we interpreted the Component C to be the primary remanence.

5.2 | Palaeogeography evolution of the Sibumasu Terrane

Previous palaeomagnetic studies on the Woniusi Formation in the Baoshan Block suggested that the Sibumasu Terrane was located at a palaeolatitude of ~42°S at the Early Permian (Ali et al., 2013; K. Huang & Opdyke, 1991; Xu et al., 2015). The Sibumasu Terrane was then

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TABLE 1 Summary of characteristic remanent magnetizations from Middle Permian limestone, Shan State of Myanmar

Site ID	Lithology	λ_s (°N)	φ_s (°E)	Strike/dip	n/n_0	Dg (°)	Ig (°)	Ds (°)	Is (°)	κ	α_{95}	φ_p (°E)	λ_p (°N)
Component A													
MT104	Dolomite limestone	20.812	97.320	145/14	6/10	358.1	38.4	5.8	30.1	50.7	9.5	226.6	82.8
MT109	Dolomite limestone	20.813	97.318	188/37	3/9	27.8	38.0	46.6	19.7	176.7	9.3	193.4	44
MT110	Dolomite limestone	20.813	97.318	148/33	6/9	18.1	25.0	22.6	−1.1	24.9	13.7	228.6	59.3
MP121	Limestone	20.985	96.700	238/23.5	6/10	355.0	34.4	7.7	54.3	69.4	8.1	121.1	74.6
MP122–123	Limestone	20.982	96.697	238/23.5	10/18	352.8	33.4	4.2	53.9	111.2	4.6	111.2	76
MK138	Limestone	21.222	96.533	170/26	4/9	342.1	47.4	10.8	44.7	25.1	18.7	157.1	78.9
MK139	Limestone	21.222	96.547	170/26	3/10	353.9	44.6	17.3	37.5	84.4	13.5	184.2	73.9
MM150 + 152	Limestone	21.894	96.223	327/18	3/20	6.0	31.6	351.0	37.7	44.5	18.7	2.7	81.6
MM153	Limestone	21.894	96.223	328/12	5/11	343.0	39.8	333.0	41.7	20.3	17.4	16.4	65.1
	Submean				**(9/9)**	**0.2**	**37.8**			**34.5**	**8.9**		
								10.3	**37.3**	**11.5**	**15.9**		
The Watson and Enkin test (1993): the optimum concentration is achieved at −15% unfolding													
Component B													
MT103	Dolomite limestone	20.813	97.321	152/41	7/10	348.7	44.4	14.3	23.7	51.4	8.5	216.7	73.9
MT105	Dolomite limestone	20.813	97.320	163/34	7/9	3.8	26.9	15.0	11.9	434.0	2.9	230.7	69.3
MT106	Dolomite limestone	20.813	97.320	164/48	8/10	351.1	31.0	12.5	16.1	15.2	14.7	231.7	72.6
MT107	Dolomite limestone	20.813	97.319	145/17	8/11	15.6	19.9	18.2	6.6	128.7	4.9	229.5	65.1
MT108	Dolomite limestone	20.813	97.319	145/17	10/13	12.5	25.1	16.3	12.4	39.3	7.8	227.6	68.5
MT111–112	Dolomite limestone	20.813	97.319	163/29	6/20	5.2	34.8	19.0	20.8	57.6	8.9	213.3	69.2
MK132 + 134	Limestone	21.198	96.560	74/23	7/20	18.0	41.0	10.0	20.4	16.1	15.5	233.0	75.7
MK135 + 136	Limestone	21.198	96.561	40/13	4/19	31.4	36.2	22.7	23.6	44.7	13.9	205.2	66.6
MK137	Limestone	21.222	96.547	170/26	6/9	348.0	13.4	354.0	13.4	69.4	8.1	299.3	74.4
	Submean				**(9/9)**	**5.9**	**31.1**			**25.5**	**10.4**		
								13.5	**16.7**	**67.6**	**6.3**		
The Watson and Enkin test (1993): the optimum concentration is achieved at 140% unfolding													
Component C													
MP118	Limestone	20.910	96.720	87/51	8/10	8.4	13.1	13.7	−27.4	134.1	4.8	254.8	52.1
MP119	Limestone	20.910	96.720	87/51	9/10	14.1	25.3	14.0	−23.9	41.3	8.1	253.1	53.9
MP120	Limestone	20.910	96.720	55/55	11/11	9.7	15.2	13.3	−24.7	266.7	2.8	254.5	53.7
	Submean				**(3/3)**	**10.7**	**17.9**			**130.3**	**10.8**	**254.1**	**53.2**
								13.7	**−25.3**	**1894.9**	**2.8**	***K* = 5181.8**	**A_{95} = 1.7**

Note. The Watson and Enkin test (1993): the optimum concentration is achieved at 108% unfolding. Abbreviations are: Dg, Ig (Ds, Is): declination and inclination of direction in situ (after tilt correction); n/n_0: number of samples used for the calculation/demagnetized; site ID: site identification; strike/dip: strike azimuth and dip of bed; κ/α_{95}: precision parameter/radius of the cone of 95% confidence for mean direction; φ_s, λ_s, latitude and longitude of sampling sites: φ_p, λ_p, latitude and longitude of corresponding virtual geomagnetic pole (VGP) after tilt correction. Numbers shown in bold indicate the submean of characteristic remanent magnetizaitons.

drifted away from the northern margin of Gondwana until it reached a palaeolatitude of ~13°S during the Middle Permian (this study). This is in good consistence with the paleontological study on reefs which suggested the eastern Cimmerian Continent (ECC) was at high latitude in the Southern Hemisphere and then arrived at low latitude near the equator (Wang & Sugiyama, 2002; Wang, Lin, et al., 2013).

Zhao et al. (2015) argued that the northern margin of the Sibumasu Terrane (Baoshan) was located at a palaeolatitude of 15.3°N in the Late Triassic and underwent a clockwise (CW) rotation of 90° after the India–Asia collision. More recently, the palaeomagnetic study on the Jurassic–Cretaceous redbed sandstones of the Tembeling Group in Malaysia Peninsular (Otofuji et al., 2017) indicates that the Malaysia Peninsula was located at a palaeolatitude of ~17°N (reference site: 4.0°N, 102.5°E) during the Jurassic and Cretaceous times, it experienced a CW rotation of 61.1° ± 11.9° with a 13.3° ± 8.1° southward displacement caused by the India–Asia collision after 55 Ma. Then, the Malaysia Peninsula experienced an additional counter clockwise (CCW) rotation of 18.5° ± 5.0° to its present location attributed to the collision between Australia and SE Asia after 30–20 Ma. The Component B of the Middle Triassic limestones indicating a CW rotation of ~120° supports this view. Besides, the palaeomagnetic data of the Jurassic–Cretaceous redbeds from Kalaw of Myanmar (Richter & Fuller, 1996) also indicate a CW rotation of ~50° relative to PGF after Palaeogene for the Shan State. However, it is worth noting that, by comparing the palaeomagnetic data of Late Triassic age (Zhao et al., 2015), Jurassic age (K. Huang & Opdyke,

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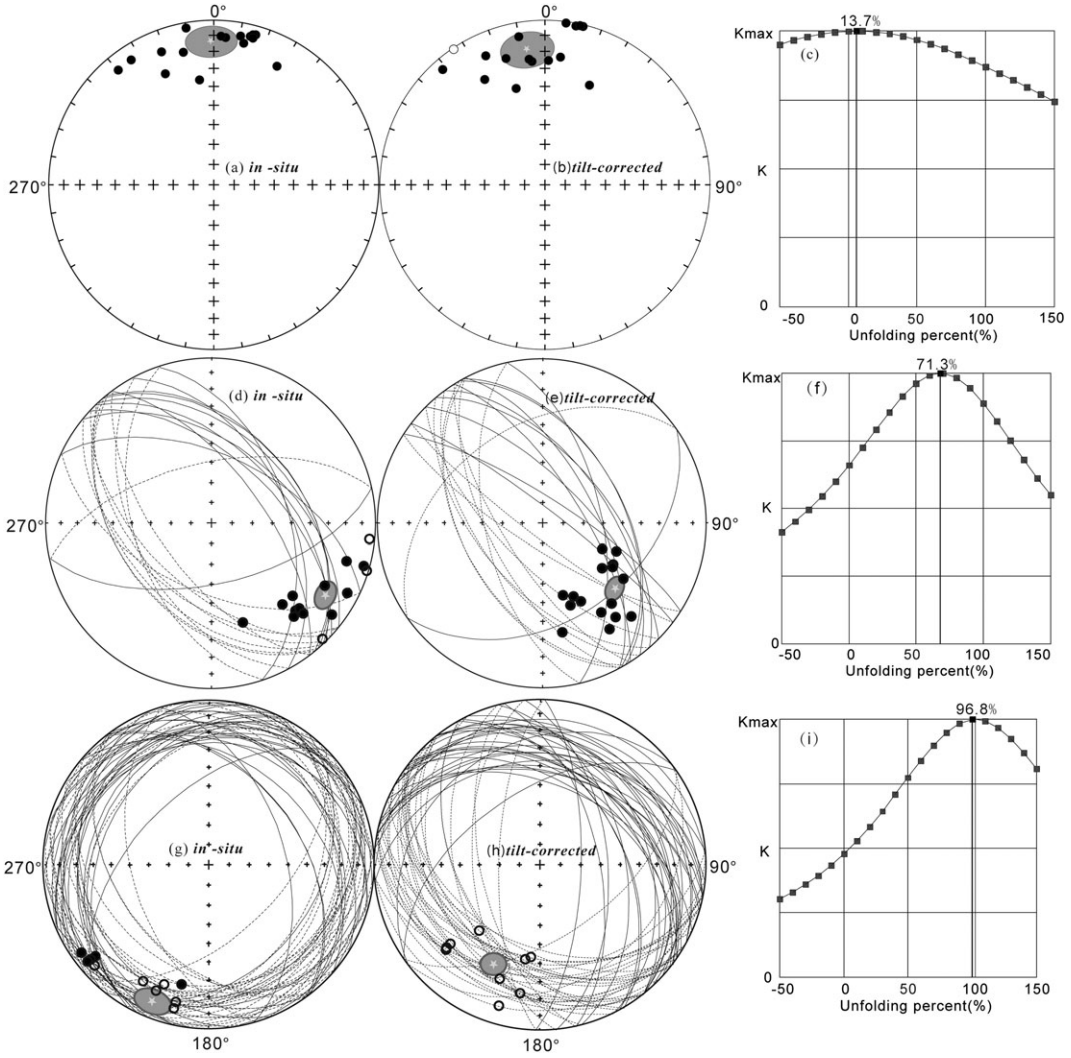


FIGURE 7 Equal-area projections of the site-mean observations from the Middle Triassic limestones from Shan State (a, d, h) before and (b, e, i) after tilt correction, (c, f, j) incremental unfolding analysis using Watson and Enkin (1993) for tilt correction, for Components A, B, C, respectively

1993), and Eocene age (Kornfeld, Eckert, Appel, Ratschbacher, Pfänder, et al., 2014) from the Baoshan Block, no obvious rotation is determined (Figure 8, Zhao et al., 2015). Based on the palaeomagnetic data from various parts of the Sibumasu Terrane, we suggested that the present N–S-trending Sibumasu Terrane is located in a narrow range of latitude between 15.0 and 18.3°N during the Middle Triassic to Jurassic. Here, we propose a tectonic model: During the Middle Triassic, the Sibumasu Terrane was located at a palaeolatitude of 15–18°N with trend of E–W and was subsequently rotated CW by ~90° as a result of the India–Asia collision. During this process, the northern margin of the Sibumasu (Baoshan) Terrane experienced a northward displacement of 9.8 ± 6.1°, corresponding to a southward displacement of 13.1 ± 6.1° (Table 2) for the southern margin (Malaysia Peninsular; Otofuji et al., 2017), whereas the central part (Shan State) does not show significant latitude displacement (this study, Table 2). All the latitude displacements are estimated with respect to its present-day geographic position.

A line of geological evidence shows that the Sibumasu Terrane was rifted from the western Australian margin of Gondwana (Metcalfe & Aung, 2014; Metcalfe, 2017). The faunas and floras of the Sibumasu Terrane are identical to those of Australia (Metcalfe, 2011). Moreover, Early Permian glacial-marine diamictites were found on the Sibumasu Terrane, indicating that the terrane must have been attached to the Himalayan-Australian Gondwana margin (Metcalfe & Aung, 2014).

On the other hand, Metcalfe (2006) conducted a palaeogeographic reconstruction of the Palaeotethys from the Permian to Triassic involving an E–W-trending subduction zone along the southern margin of the Indochina Block, which is further supported by the presence of mélange and thrust structures derived from illite crystallinity data (Hara et al., 2009). Collectively, we infer that the Sibumasu Terrane trended E–W during the Permian to Jurassic times.

5.3 | The closure of the eastern Palaeotethys Ocean

Reliable palaeomagnetic results suggest that the Indochina Terrane was located at a palaeolatitude of 21.2 ± 5.7°S during the Early–Middle Permian (Yan, Huang, Zhang, Charusiri, & Veeravinantanakul, 2018),

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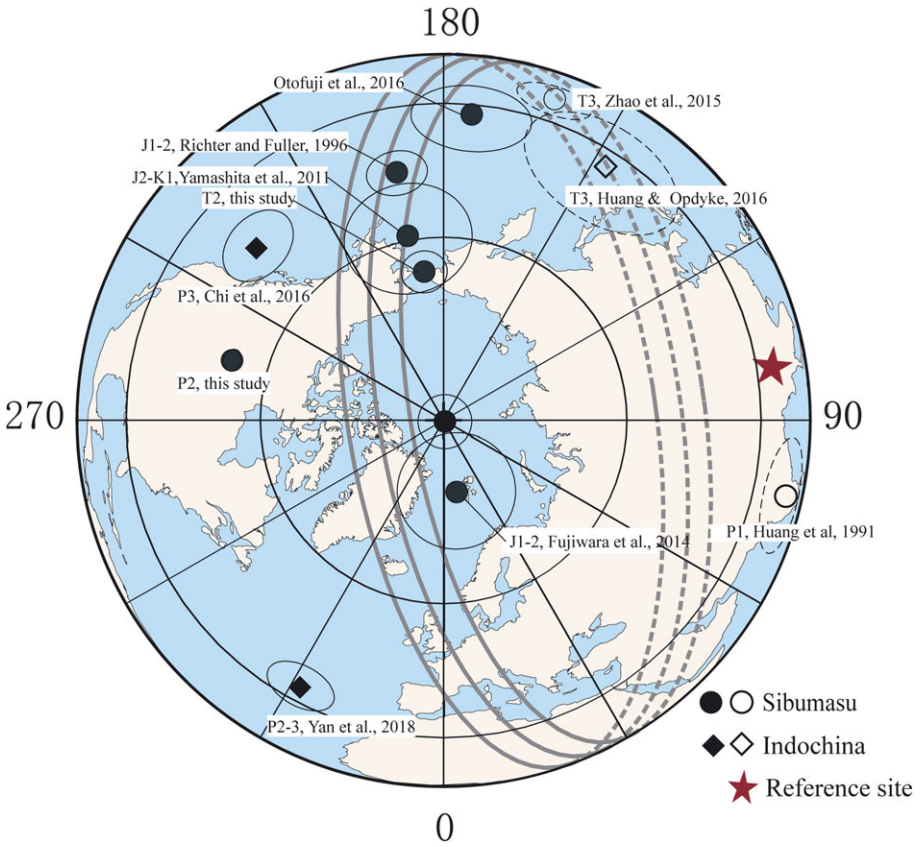


FIGURE 8 Orthographic projection showing the palaeomagnetic poles derived from the Middle and Middle Triassic limestone in Shan State (this study) in comparison with the Upper Triassic Yiwanshui Formation (T3y) in the Simao Terrane (K. Huang & Opdyke, 2016) and Niuhetang Formation (T3n) in the Baoshan Terrane (Zhao et al., 2015). P1: Early Permian; P2: Middle Permian; P3: Late Permian; T2: Middle Triassic; T3: Late Triassic; J1-2: Early-Middle Jurassic; J2-K1: Middle Jurassic to Early Cretaceous [Colour figure can be viewed at wileyonlinelibrary.com]

and it arrived at 15.6 ± 13.5°N (reference site: 23.9°N, 99.2°E) in the Late Triassic (K. Huang & Opdyke, 2016), whereas the Baoshan Block was located at ~42°S (Ali et al., 2013; K. Huang & Opdyke, 1991; Xu et al., 2015) in the Early Permian, it then drifted northward from Gondwana and reached a palaeolatitude of 15–18°N in the Late Triassic (Figure 9; Zhao et al., 2015). The Component C from the Middle Triassic limestones indicates a palaeolatitude of ~18.3°N (Table 2, Figure 8). As shown in Figure 8, all the Middle–Late Triassic palaeomagnetic poles from the Sibumasu Terrane and Indochina Terrane are located around a small circle centred at the same reference sites, we infer that these two terranes are located at a similar latitude in the Northern Hemisphere during the Middle–Late Triassic, and the Sibumasu–Indochina collision occurred before or during this period of time.The Sibumasu and Indochina collision before the Middle–Late Triassic is demonstrated by sufficient geological evidence. The Inthanon Suture, separating the Sibumasu Terrane from the Sukhothai Arc and Indochina Terrane, comprises Permian basaltic volcanoes (Barr, Tantisukrit, Yaowanoiyothin, & Macdonald, 1990; Phajuy, Panjasawatwong, & Osataporn, 2005), Early Carboniferous–Late Permian limestones (Metcalfe, 2002; Vachard, Fontaine, & Caridroit, 1992) and Devonian–Triassic radiolarian cherts (Caridroit, 1993; Sashida et al., 2000; Sashida, Igo, Hisafa, Nakornsri, & Ampornmaha, 1993; Sashida, Salyapongse, & Nakornsri, 2000; Wonganan & Caridroit, 2005). This is correlated with the Changning–Menglian Suture in Yunnan, which is made up of ophiolitic mélanges, volcanics, shallow-marine carbonates, and deep-marine sedimentary sequence of Devonian–Middle Triassic age (Feng, 2002; Feng & Liu, 1993; Feng & Ye, 1996; He & Liu, 1993; Sone & Metcalfe, 2008; Ueno, Wang, & Wang, 2003; Zhao, Zhong, & Wang, 1994). Furthermore, the granites, formed at 230–220 Ma in the Inthanon and Sukhothati belt, are interpreted as the post-collisional products (Y. Wang et al., 2016), indicative of a collision before the Mid-Triassic. This can be also correlated to the post-collisional Lincang granite (~230–220 Ma; Dong et al., 2013; Peng, Wilde, Wang, Fan, & Peng, 2013; Y. Wang et al., 2010). Additionally, the Chiang Khong–Lampang volcanic rocks in the NW Thailand representing two age ranges of 242–237 and 230–220 Ma are compatible with the Manghuai and Xiaodingxi/Manghuihe volcanic sequences in the SW Yunnan province of China that are interpreted as syn- and post-collisional products (Barr, Macdonald, Ounchanum, & Hamilton, 2006; Panjasawatwong, 2003; Qian, Feng, Chonglakmani, & Monjai, 2013; Qian et al., 2016a, 2016b; Srichan, Crawford, & Berry, 2009; Y. Wang et al., 2010). Besides, these volcanic rocks overlie uncomfortably on the Permian–Lower Triassic and are overlain by the Upper Triassic–Lower Jurassic molasses demonstrates that the collision of the Sibumasu Terrane and Indochina Terrane ended by the Late Triassic (Hara et al., 2009; Ridd, Barber, & Crow, 2011). Last but not least, during the Early Permian (Asselian–Sakmarian), faunas of the

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TABLE 2 Reliable Permian to Cretaceous palaeomagnetic results available from Sibumasu Terrane and the Indochina Terrane

Locality name	Lat. (°N)	Long. (°E)	Age	S/N		Fold test	Observed direction			Palaeomagnetic pole			Pla. (°N)	Lat. DP (°)	Reference
							Dec. (°)	Inc (°)	α_{95} (°)	Lat. (°N)	Lon. (°E)	A_{95} (°)			
Sibumasu															
North Trang syncline	8.1	99.4	J2-K2	9	Redbeds	+	31.1	12.2	13.9	59.1	190.9	10.1	6.2	1.9 ± 10.1°	Yamashita et al. (2011)
Peninsular Malaysia	4.0	102.5	J3-K1	22	Redbeds	+	56.8	31.6	11.3	32.9	174.7	12.7/7.1	17.1	−13.1 ± 9.5°	Otofuji et al. (2016)
Kalaw, Myanmar	20.7	96.5	J-K	13	Redbeds	+	44.7	23.4	6.1	46.4	190.6	4.8	12.2	8.5 ± 4.8°	Richter and Fuller (1996)
Ratchaburi	13.5	99.7	J1-J2	19	Redbeds	+	348.5	24.7	10.5	78.6	10.6	9.3	13	0.5 ± 3°	Fujiwara et al. (2014)
Baoshan	23.9	99.2	T3	15/	Basalts	+	122.8	26.9	6.6	−21.7	160.9	6.1	14.2	9.8 ± 6.1°	Zhao et al. (2015)
Shan State	20.2–21.2	96.3–96.7	T2	–	Limestone	+	205.6	−33.5	4.2	65.8	187.3	2.7/4.8	18.3	2.3 ± 3.6°	This study
Shan State	20.2–21.2	96.3–96.7	P2	3/28	Limestone	+	13.7	25.3	2.8	53.2	254.1	1.7	−13.0	–	This study
Baoshan	25.2	99.3	P1	13/65	Basalts	+	209.2	61.0	6.7	−17.2	77.5	9.3	−42.0	–	Huang and Opdyke (1991)
Indochina															
Simao, Yunnan	23.43–23.67	101.67–101.76	T3	12/	Redbeds	+	414.4	31.2	13.0	−34.8	147.5	13.5	15.6		K. Huang and Opdyke (2016)
Thuan Chau, Vietnam	21.38	103.73	P3	8/74	Basalt	–	216.1	10.5	8.9	45.6	226.8	4.6/9.0	−5.8		Chi, Geissman, Quy, Anh, and Dung (2016)
Saraburi	15.82–14.83	101.35–100.93	P1–2	15/105	Limestone	+	136.3	37.2	6.1	34.1	331.7	5.7	−13.5		Yan et al. (2018)

Note. Lat. and Long. refer to latitude and longitude, respectively. The ages are as follows: T2 is Middle Triassic, T3 is Late Triassic, J is Jurassic, J1 is Early Jurassic, J2 is Middle Jurassic, J3 is Late Jurassic; K is Cretaceous, K1 is Early Cretaceous, K2 is Late Cretaceous. N/n is the number of sites/samples used for palaeomagnetic calculations. Dec. and Inc. refer to declination and inclination, respectively, α_{95}/A_{95} is the radius of the cone at the 95% confidence level with respect to the mean direction/palaeomagnetic pole. Pal. refer to palaeolatitude. Lat. DP refer to latitude displacement relative to present location, positive and negative values represent northward and southward transition.

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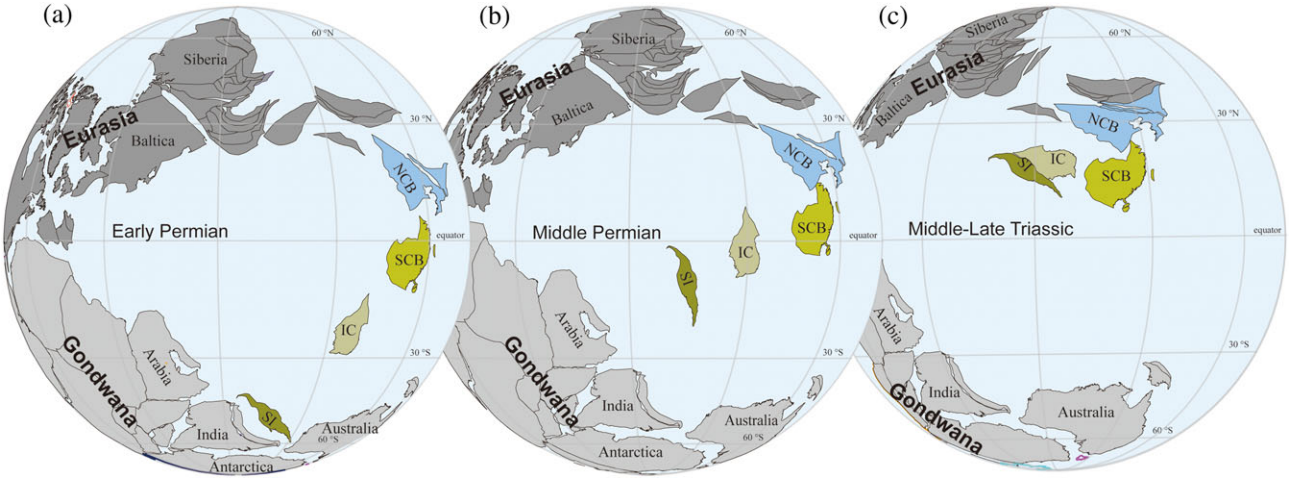


FIGURE 9 Reconstruction of the Sibumasu Terrane (SI), Indochina Terrane (IC), South China Block (SCB), and North China Block (NCB) in the Pangea supercontinent at Early Permian (a), Middle Permian (b), and Middle–Late Triassic (c). Data used are included in Table 2. (after B. Huang et al., 2018) [Colour figure can be viewed at wileyonlinelibrary.com]

Sibumasu Terrane were sourced by the peri-Gondwana Indoralian province. As the Sibumasu Terrane was rifted away from Gondwana and moved northwards during the Permian, its faunal characteristics changed initially to endemic Sibumasu Province in the Middle Permian and subsequently to Cathaysian Province faunas in the Late Permian (Metcalfe, 2011; Shi & Archbold, 1998; Ueno, 2003). Based on the evidence presented above, our results support the view that the tectonic collision between the Sibumasu Terrane and the Indochina Terrane occurred during or before the Middle–Late Triassic.

6 | CONCLUSIONS

We presented a detailed palaeomagnetic study on the Middle Permian and Middle Triassic limestones from the Shan State of the Sibumasu Terrane in eastern Myanmar. Our new results highlight three stable ChRM components. The Component C of Middle Permian age passed the fold test at 95% confidence level and therefore represented the primary origin. It has been acquired during or shortly after rock formation. This suggests that the Sibumasu Terrane is located at ~13°S during the Middle Permian. The Component C separated from the Middle Triassic limestones also passed the fold test and is of primary origin, defining a palaeolatitude of ~18.3°N during the Middle Triassic. This palaeomagnetic evidence supports the view that the collision between the Sibumasu Terrane and Indochina Terrane was completed during or before the Middle–Late Triassic.

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